

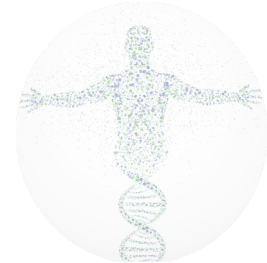


Ecole Thématique #2 Métabolomique et Santé Humaine

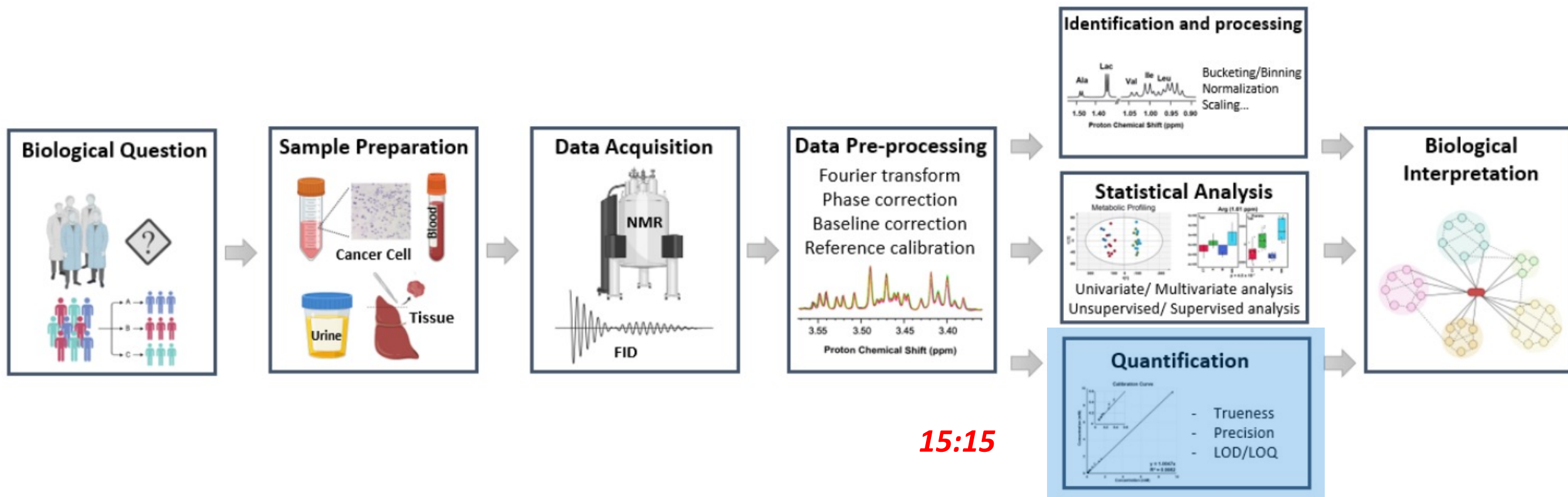
Aussois, 5 - 9 octobre 2026



SFBC
Société Française
de Biologie Clinique



NMR Session



15:15

Corentine Goossens, Gildas Bertho

Metabolomics & Human Health



October 5-9, 2026

Paul Langevin Center CAES CNRS, Aussois, Savoie

**FREE OF
CHARGE**

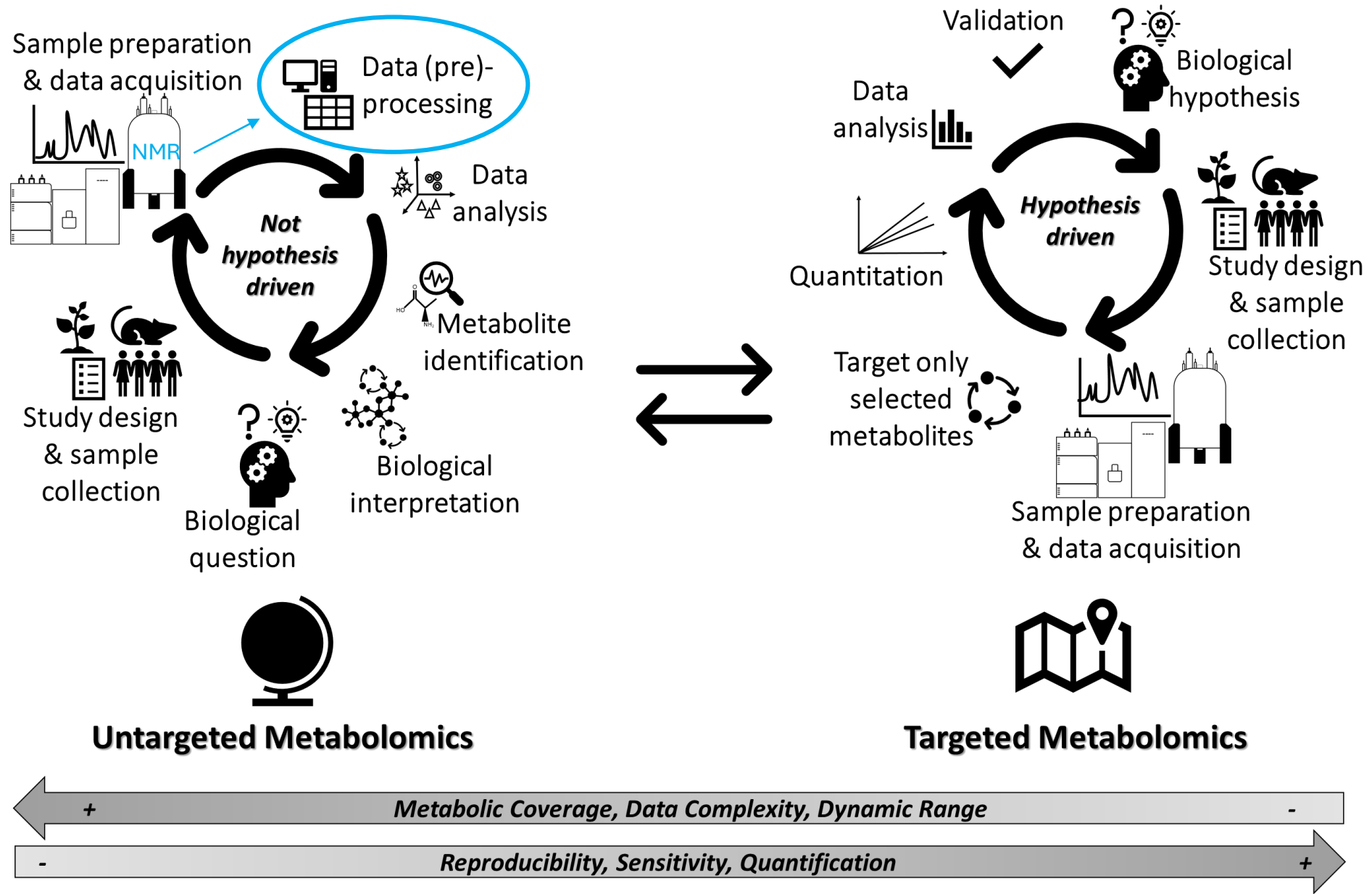
FOR CNRS EMPLOYEES

Session NMR-Wednesday morning

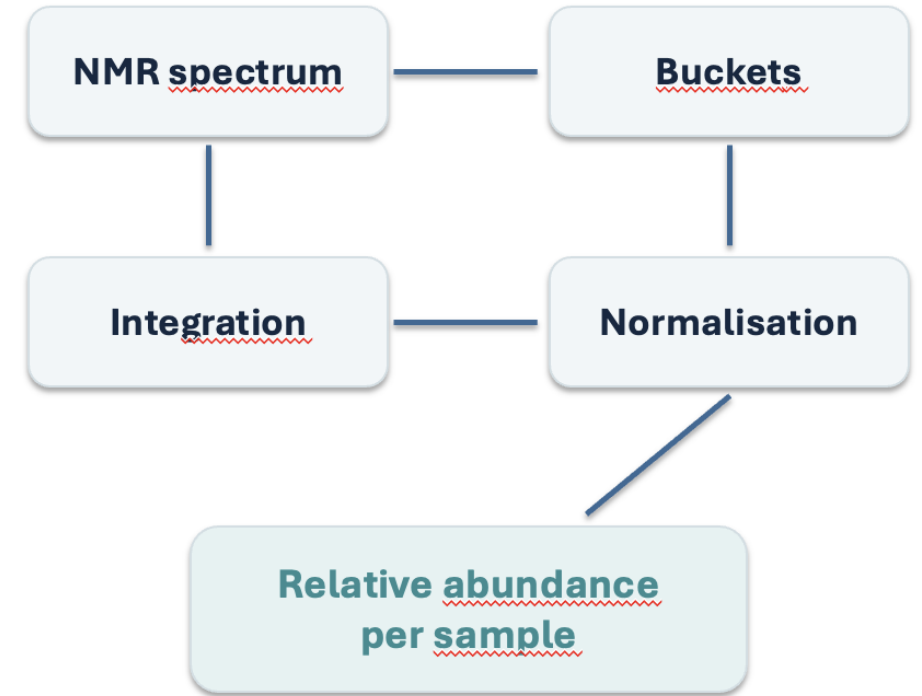
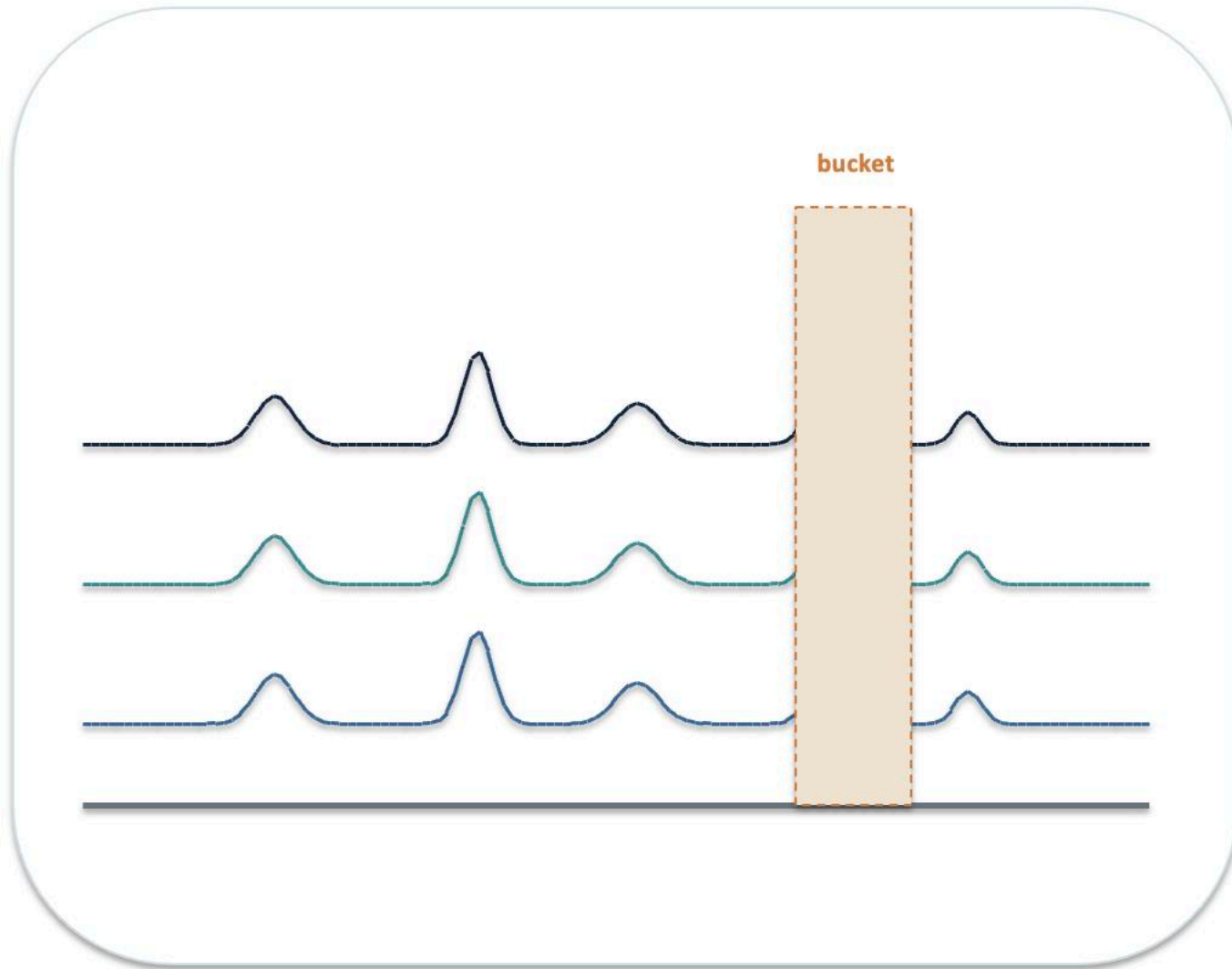
Workshop 4 : NMR-based quantification in clinical metabolomics

Corentine GOOSSENS ; Gildas Bertho

Metabolomics – *Untargeted NMR approach*



Untargeted NMR approach : « relative quantification »



Objective : Bucket comparison

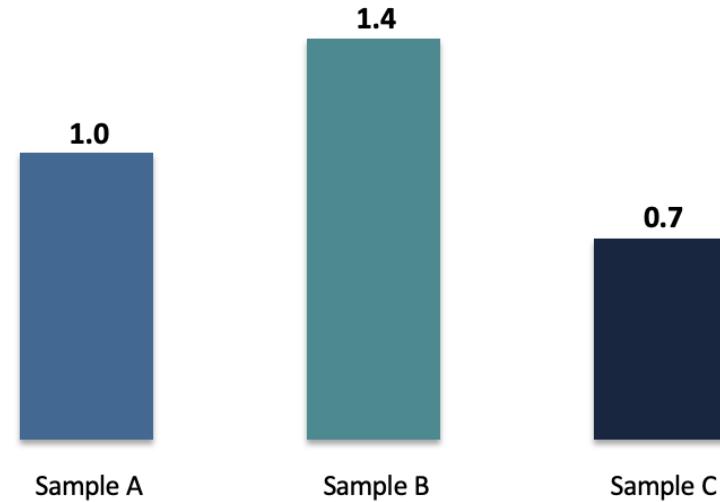
Untargeted NMR approach : « relative quantification »

= a comparable measure of intensity but not an absolute concentration

$$X_{ij} = \int S_i(\delta) d\delta$$

X_{ij} : area under the signal / bucket j in sample i
The area reflects the signal abundance under comparable acquisition conditions

Exemple : relative abundance



RELATIVE
comparison / fold-change / statistics

ABSOLUTE
concentration : $\mu\text{mol/L}$, mmol/L ...

Untargeted NMR approach : « relative quantification »



Importance of normalisation

A difference between integrals is not necessarily a biological difference

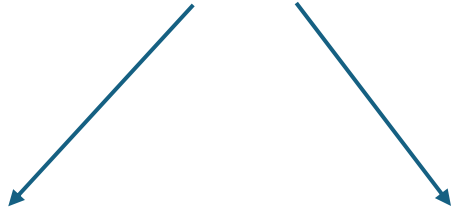
Observed variation = biology + dilution + analytical factors

Metabolomics – Targeted NMR approach

-> a set of known metabolites are quantitated



internal or external reference compounds



available databases

using standard compounds

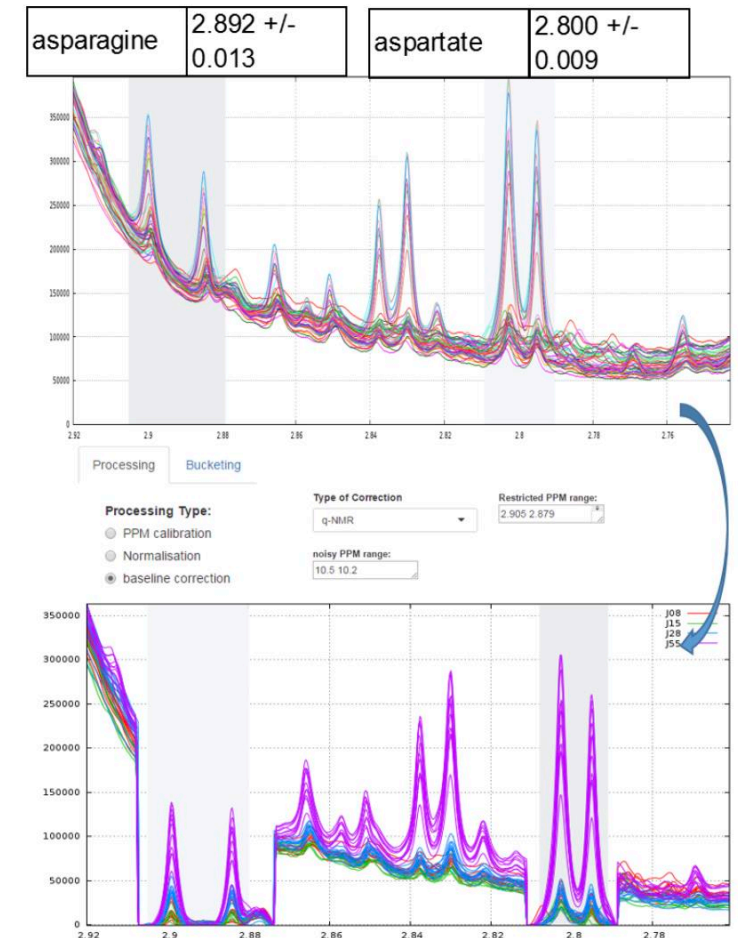
-> Correction of the baseline :

- 1) eliminate the residual effects due to the presence of macromolecules in extracts
- 2) reduce the prevalence of a high intense peak on the less intense ones



when patterns are not overlapped !

q-HNMR baseline correction



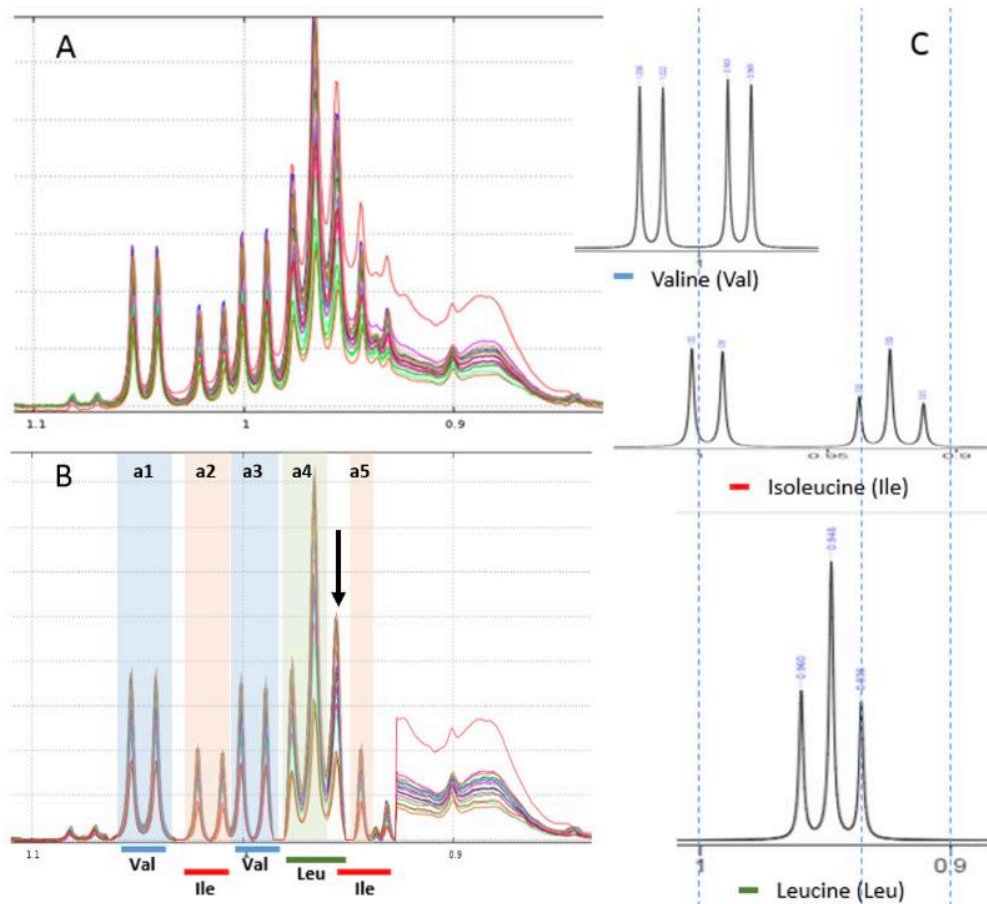
Metabolomics – Targeted NMR approach

Calibration-curve method

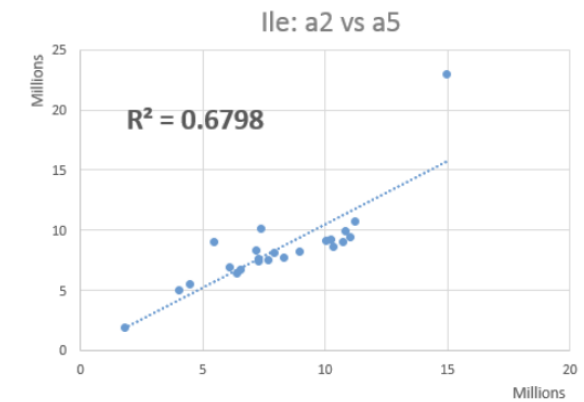
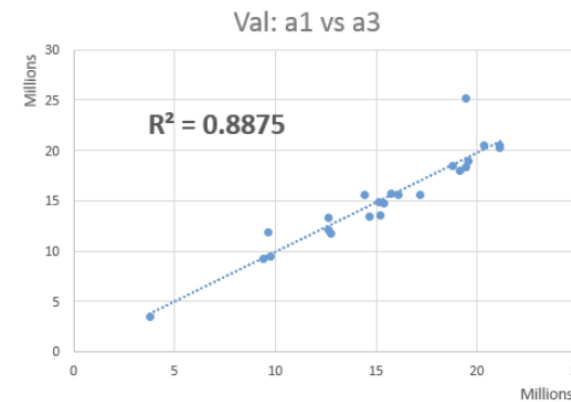
(i.e. external reference or/and)

Quality of the baseline correction

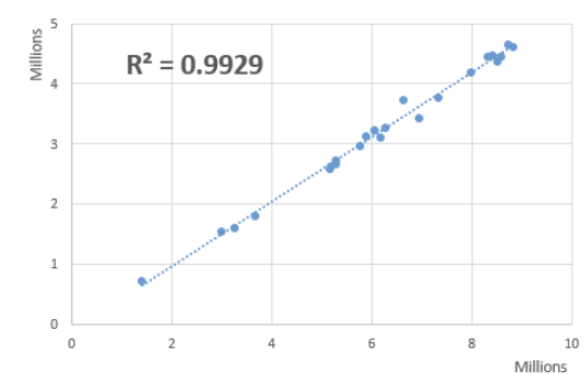
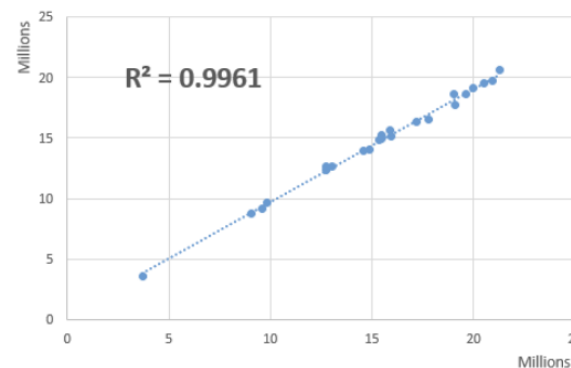
Correlation between 2 buckets belonging to the same compound, before and after the baseline correction



Before



After



Metabolomics – *Targeted NMR approach*

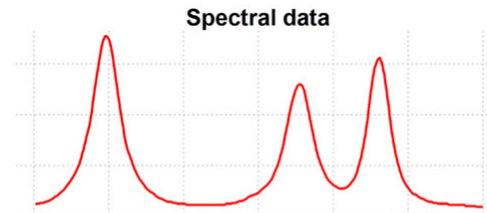
In crowded regions with heavy peak overlap -> deconvolution methods

- Bayesian Quantification (*BQuant*)

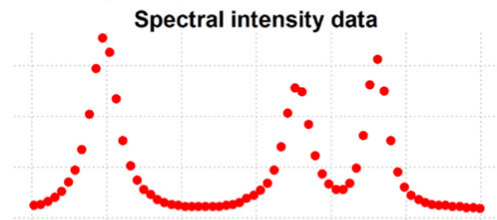
Information & Extraction of data

Different types of features can be extracted from spectral data and used for data analysis

- ✓ Peak position: frequencies obtained after Fourier transformation
- ✓ Peak integral: related to quantity

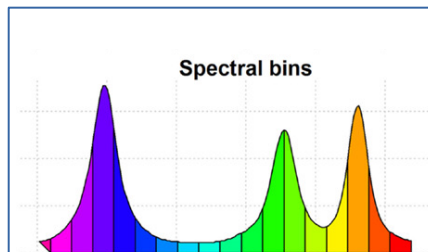


Full resolution

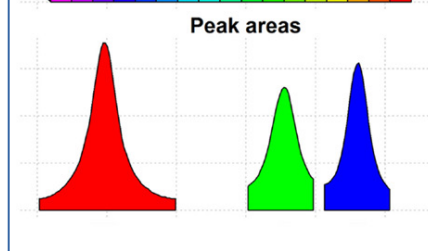


Bining = bucketing

Uniform

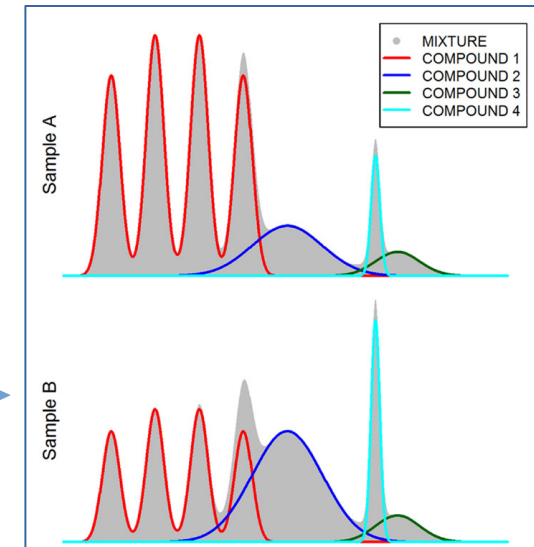


Intelligent



- Older method, largely used
- Overlapping not managed

Deconvolution

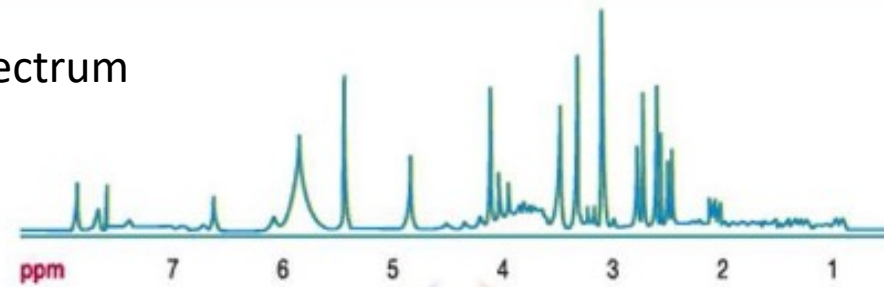


- Rather good quantification method but needs a complete database (€€€€€€) !

Workflow

Measurements

NMR Spectrum



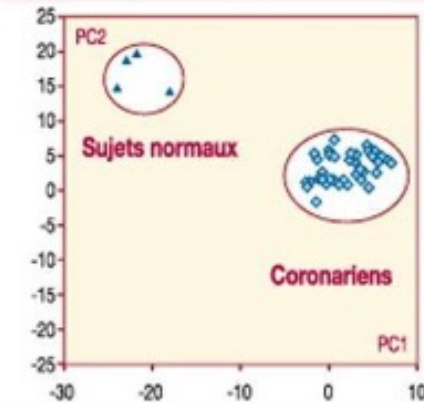
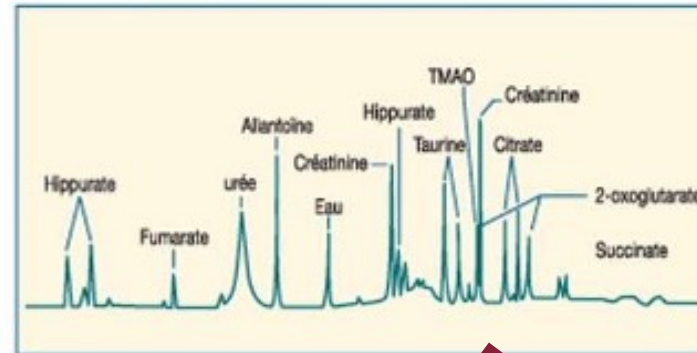
Targeted analysis
(main metabolites)

Untargeted approach
(all metabolites)

Méthode quantitative
(reconnaissance de signatures de métabolites)

Méthode chimiométrique
(reconnaissance de signatures de maladies)

Datasets



Statistical analysis

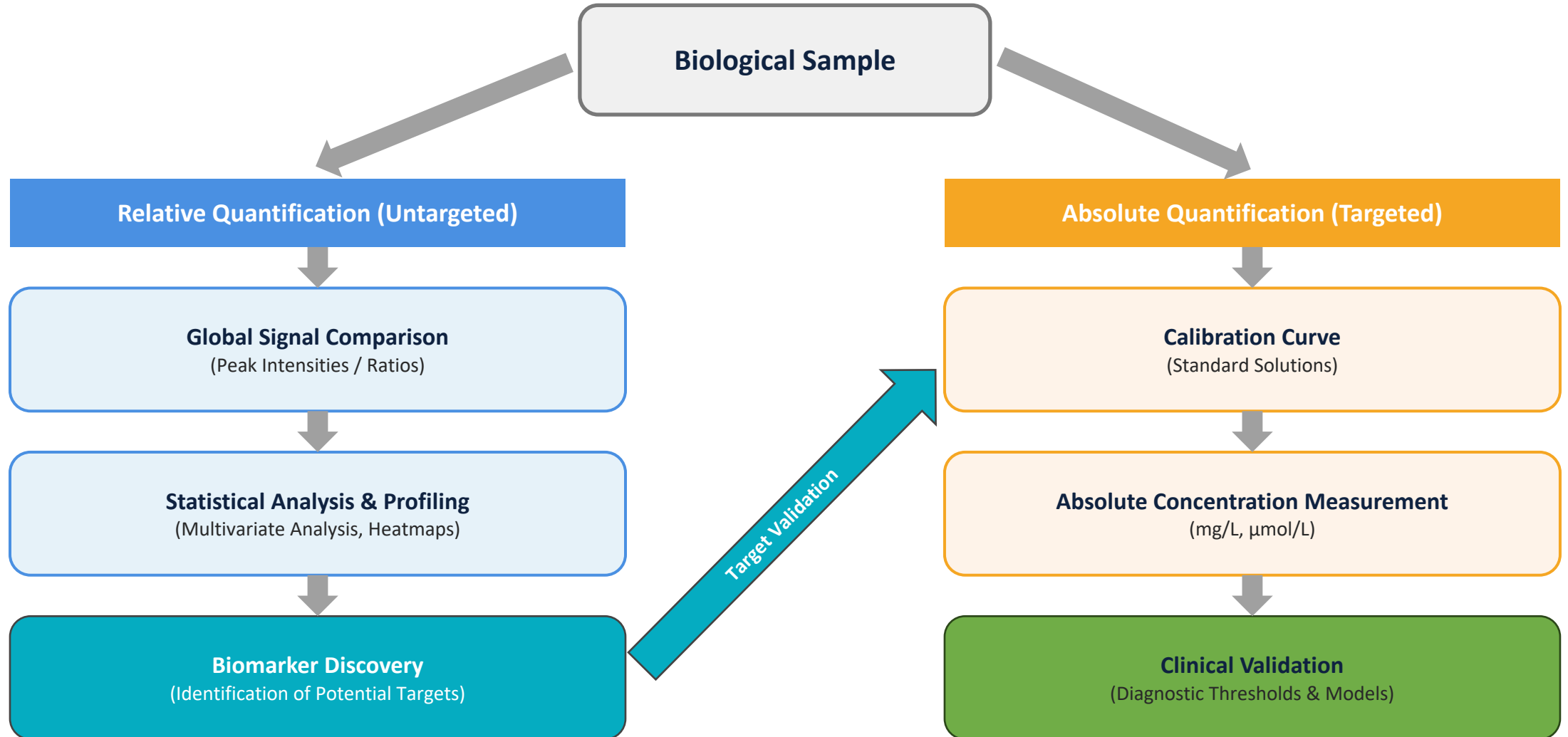
IVDr report

Uni- & Multi-variate analysis, Integration with metadata, Interpretation

Relative vs Absolute Quantification ?

Metabolomics Quantification Workflow

Integrating Relative and Absolute Strategies (Without Isotopic References)

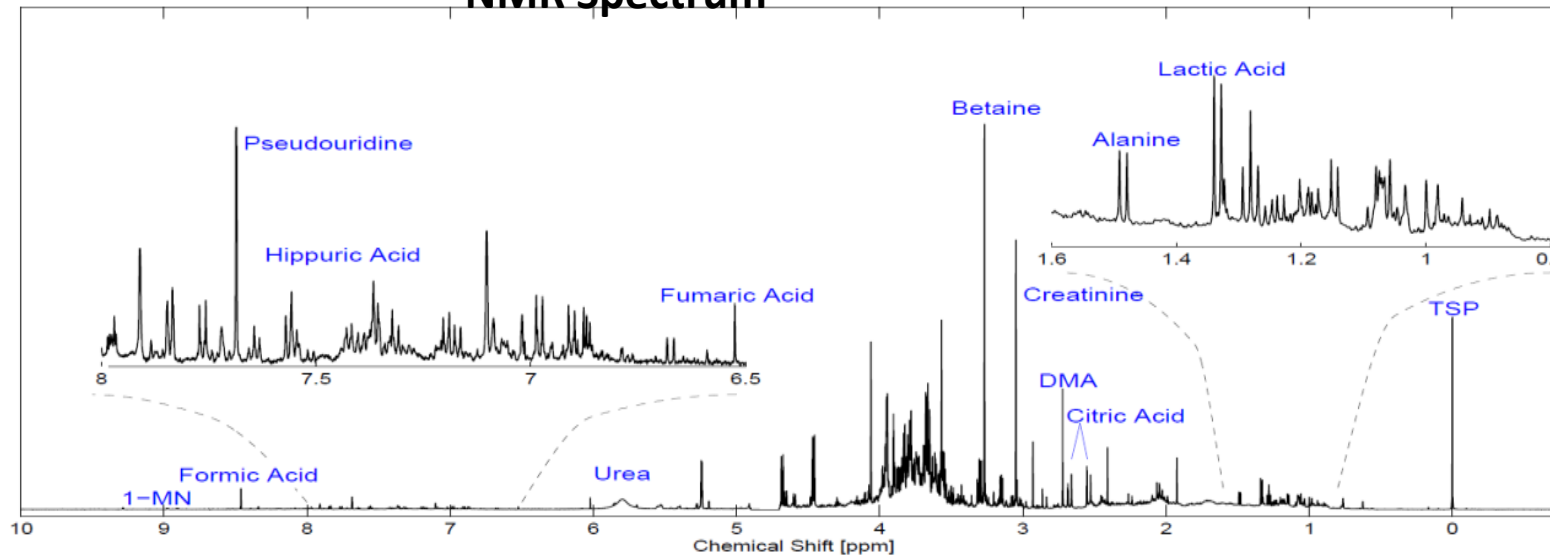


TARGETED ANALYSIS

Automatic quantification of metabolites

Urines

NMR Spectrum



➔ Analysis report (illustration part)

5 Benzene and substituted derivatives

Compound	Conc.	Conc.	LOD	r	p	Δ	95% Range ^(*)
	mmol/L	mmol/L	mmol/L			mmol/L	mmol/L
2-Hydroxyphenylacetic acid	< 0.01	< 10	10	0.000	0	0.404	< 10
3-Phenylactic acid	< 0.07	< 89	89	0.040	0	0.268	< 89
4-Aminohippuric acid	< 0.21	< 270	270	0.000	0	0.270	< 270
4-Ethylphenol	< 0.01	< 13	13	0.000	0	0.290	< 13
4-Hydroxyhippuric acid	0.20	260	26	0.195	82	0.070	< 30
4-Hydroxyphenylacetic acid	0.06	82	18	0.062	95	0.049	< 28
4-Hydroxyphenyllactic acid	< 0.03	< 43	43	0.000	0	0.608	< 43
Benzoic acid	0.16	210	10	0.158	99	0.013	< 10
D-Mandelic acid	< 0.01	< 2	2	0.000	0	0.100	< 2
Hippuric acid	0.38	500	170	0.380	94	0.079	< 660
Phenylacetic acid	< 0.06	< 72	72	0.017	80	0.095	< 72
Phenylpyruvic acid	0.09	110	98	0.085	98	0.016	< 98
Pyrocatechol	< 0.13	< 170	170	0.000	0	0.312	< 170
Syringic acid	0.18	230	38	0.177	99	0.075	< 38

^(*) Gray horizontal boxes represent 95% concentration range, black vertical lines represent sample value.

6 Carboxylic acids

Compound	Conc.	Conc.	LOD	r	p	Δ	95% Range ^(*)
	mmol/L	mmol/L	mmol/L			mmol/L	mmol/L
5-Aminolevulinic acid	< 0.01	< 2	2	0.000	0	0.253	< 2
Acetic acid	0.48	630	5	0.478	100	0.028	< 51
Citric acid	2.4	3100	40	2.352	100	0.234	< 700
E-Glutaconic acid	< 0.03	< 38	38	0.000	0	0.055	< 38
Ethylmalonic acid	0.06	78	35	0.059	65	0.067	< 35
Formic acid	0.09	120	10	0.092	100	0.003	< 43
Fumaric acid	0.00	3	2	0.002	70	0.001	< 3
Glutaric acid	< 0.13	< 170	170	0.020	0	0.171	< 170
Imidazole	0.05	61	48	0.046	86	0.024	< 48
Lactic acid	0.38	500	49	0.377	99	0.176	< 110
Maleic acid	< 0.01	< 4	4	0.003	87	0.001	< 8
Methylmalonic acid	< 0.01	< 17	17	0.000	0	0.287	< 31
Propionic acid	< 0.04	< 51	51	0.000	0	0.079	< 51
Succinic acid	0.11	140	5	0.108	96	0.021	< 39
Tartaric acid	0.09	120	5	0.090	100	0.009	< 110
Trigonelline	< 0.03	< 35	35	0.013	89	0.003	< 67
Xanthurenic acid	< 0.01	< 18	18	0.000	0	0.075	< 18

^(*) Gray horizontal boxes represent 95% concentration range, black vertical lines represent sample value.

➤ Up to 150 metabolites in 15 min

(Up to 1000 compounds are detected)

➤ Newborn screening : 75 IEM (inherited errors of metabolism)

TARGETED ANALYSIS

Automatic quantification of metabolites

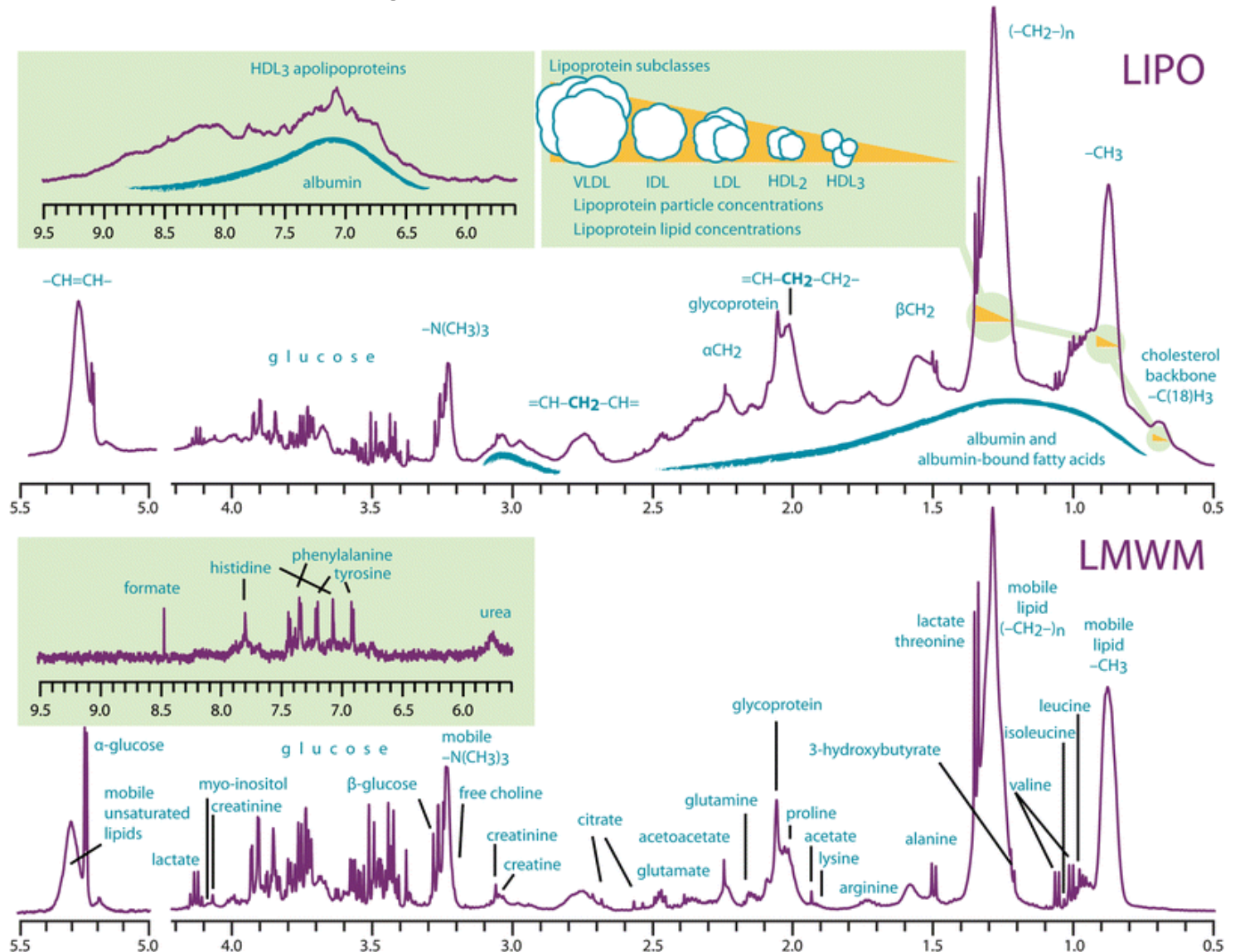
BLOOD

NMR Spectrum

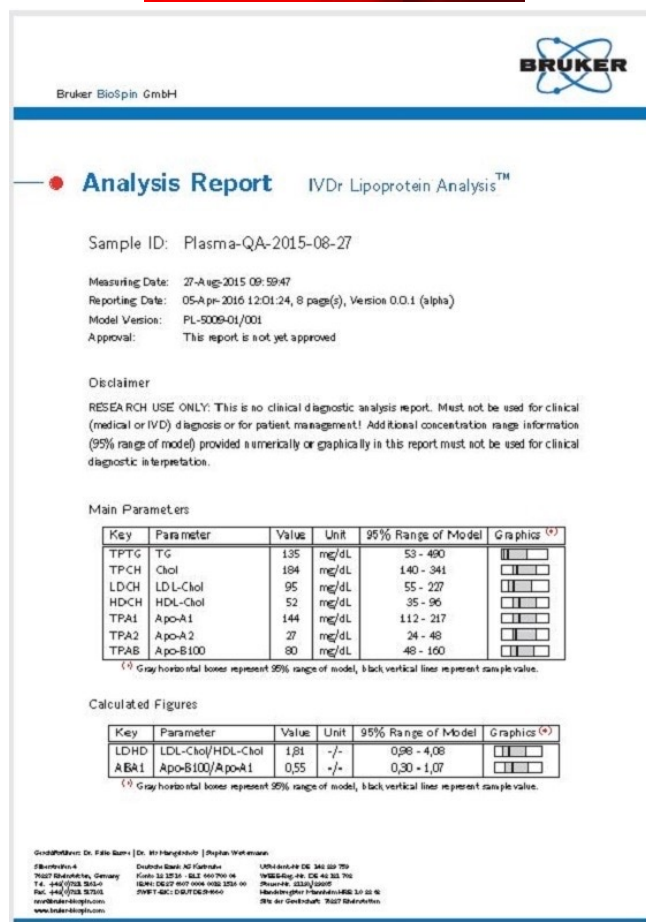
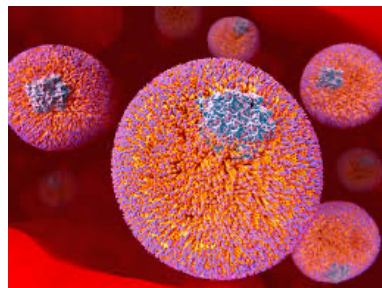
2 profiles :

➤ Lipoprotein profiles :
112 parameters

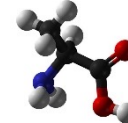
➤ Metabolic profiles :
41 parameters



Lipoprotein report



Metabolite report



Brüker BioSpin GmbH



Analysis Report

Brüker IVD_r Quantification in Plasma/Serum B.I. Quant-PS™

Sample ID: I004.S001.000386

Measuring Date: 08-Apr-2009 13:05:12
Reporting Date: 29-Jul-2019 17:16:24, 7 page(s), Version 2.0.0
Quantification Method Version: Quant-PS 2.0.0

Disclaimer

RESEARCH USE ONLY: This is no clinical diagnostic analysis report. Must not be used for clinical (medical or IVD) diagnosis or for patient management! Additional concentration range information (95% range) provided numerically or graphically in this report must not be used for clinical diagnostic interpretation.

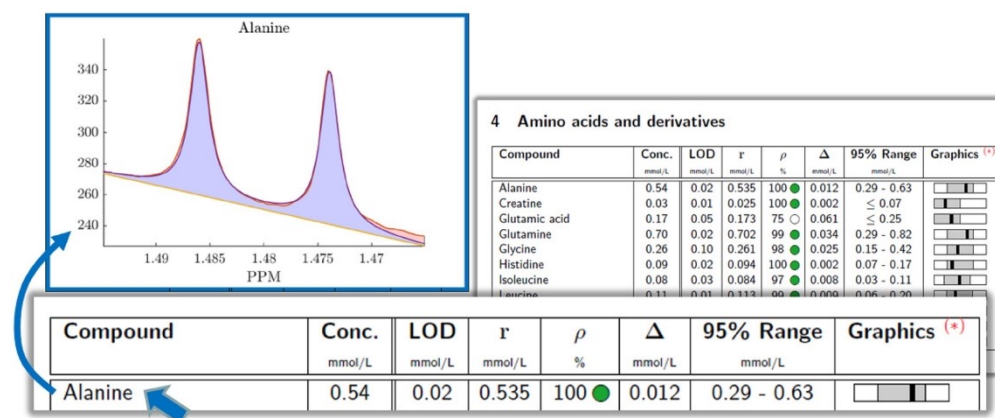
Application of B.I. Quant-PS 2.0 requires use of Brüker's B.I. Methods SOP for plasma and serum.

Summary

The spectroscopic fingerprint of the sample is consistent with a serum or a heparin plasma profile. The following metabolites were found with concentrations outside the 95% range of Brüker Quant-PS 2.0.0 plasma/serum metabolite concentration data base:

Amines and derivatives: Trimethylamine-N-oxide (2.54 mmol/L).
Amino acids and derivatives: Phenylalanine (0.07 mmol/L).
Carboxylic acids: Formic acid (0.07 mmol/L), Succinic acid (0.01 mmol/L).
Keto acids and derivatives: Pyruvic acid (0.09 mmol/L).

Further detailed information is provided on the following pages.



by posing the mouse cursor on the metabolite name, the corresponding graphical fit display pops up in PDF report

List of parameters (targeted approach)

Lipoprotein report

TPTG	HDA2	L5CH	H3PL
TPCH	VLAB	L6CH	H4PL
LDCH	IDAB	L1FC	H1A1
HDCH	LDAB	L2FC	H2A1
TPA1	V1TG	L3FC	H3A1
TPA2	V2TG	L4FC	H4A1
TPAB	V3TG	L5FC	H1A2
TPTG	V4TG	L6FC	H2A2
ABA1	V5TG	L1PL	H3A2
TBPN	V1CH	L2PL	H4A2
VLPN	V2CH	L3PL	
IDPN	V3CH	L4PL	
LDPN	V4CH	L5PL	
L1PN	V5CH	L6PL	
L2PN	V1FC	L1AB	
L3PN	V2FC	L2AB	
L4PN	V3FC	L3AB	
L5PN	V4FC	L4AB	
L6PN	V5FC	L5AB	
VLTG	V1PL	L6AB	
IDTG	V2PL	H1TG	
LDTG	V3PL	H2TG	
HDTG	V4PL	H3TG	
VLCH	V5PL	H4TG	
IDCH	L1TG	H1CH	
VLFC	L2TG	H2CH	
IDFC	L3TG	H3CH	
LDFC	L4TG	H4CH	
HDFC	L5TG	H1FC	
VLPL	L6TG	H2FC	
IDPL	L1CH	H3FC	
LDPL	L2CH	H4FC	
HDPL	L3CH	H1PL	
HDA1	L4CH	H2PL	

Metabolite report

Ethanol	2-Oxoglutaric acid
Trimethylamine-N-oxide	3-Hydroxybutyric acid
2-Aminobutyric acid	Acetoacetic acid
Alanine	Acetone
Asparagine	Pyruvic acid
Creatine	D-Galactose
Creatinine	Glucose
Glutamic acid	Glycerol
Glutamine	Dimethylsulfone
Glycine	
Histidine	Ca-EDTA
Isoleucine	K-EDTA
Leucine	
Lysine	
Methionine	
N,N-Dimethylglycine	
Ornithine	
Phenylalanine	
Proline	
Sarcosine	
Threonine	
Tyrosine	
Valine	
2-Hydroxybutyric acid	
Acetic acid	
Citric acid	
Formic acid	
Lactic acid	
Succinic acid	
Choline	



Lipid parameters

Biological lab

4 parameters

IVDr system

112 parameters !

LIPID PROFILE			
	DESIRABLE	BORDERLINE	HIGH RISK
Cholesterol	<200 mg/dl	200-239 mg/dl	240 mg/dl
Triglycerides	<150 mg/dl	150-199 mg/dl	200-499 mg/dl
HDL cholesterol	60 mg/dl	35-45 mg/dl	<35 mg/dl
LDL cholesterol	60-130 mg/dl	130-159 mg/dl	160-189 mg/dl
Cholesterol/HDL ratio	4.0	5.0	6.0

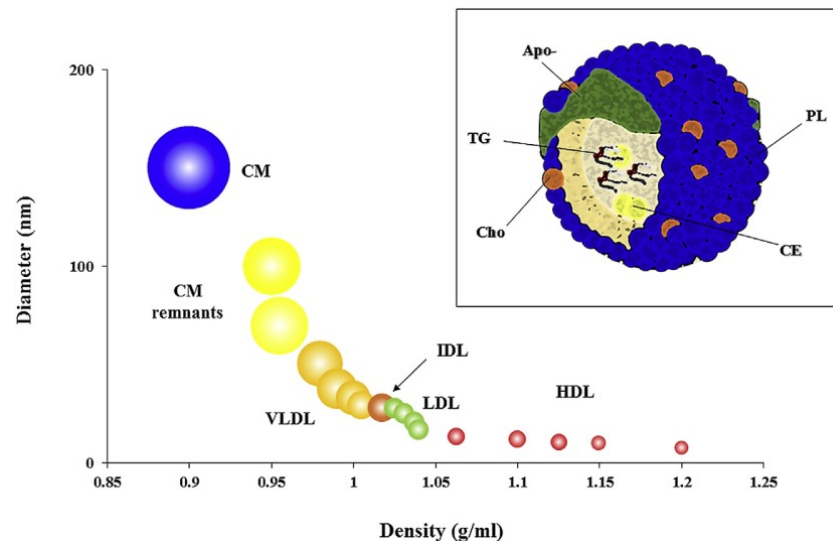
Main Parameters

Key	Parameter	Value	Unit
TPTG	TG	146	mg/dL
TPCH	Chol	172	mg/dL
LDCH	LDL-Chol	85	mg/dL
HDCH	HDL-Chol	50	mg/dL
TPA1	Apo-A1	133	mg/dL
TPA2	Apo-A2	30	mg/dL
TPAB	Apo-B100	68	mg/dL

Possible Filters

95% Range of Model	Graphics (*)
53 - 490	
140 - 341	
55 - 227	
35 - 96	
112 - 217	
24 - 48	
48 - 160	

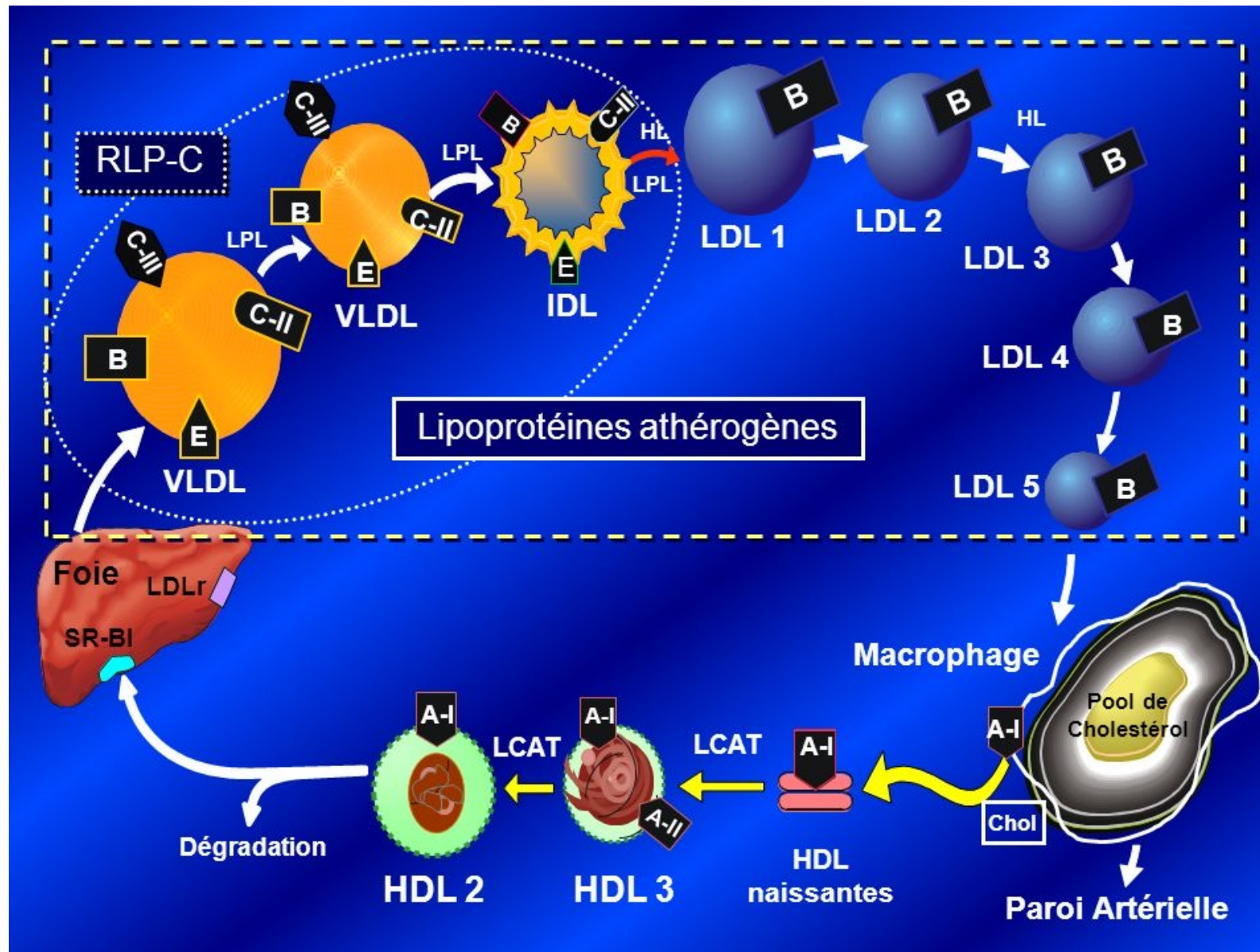
(*) Gray horizontal boxes represent 95% range of model, black vertical lines represent sample value.



Additional parameters

	Triglyceride	Cholesterol	Free Cholesterol	Phospho-lipids	Apo-A1	Apo-A2	Apo-B	LDL-particle
PLASMA	✓	✓	✓	-	✓	✓	✓	✓
VLDL	✓	✓	✓	✓			✓	✓
VLDL-1 to -6	✓	✓	✓	✓	-	-	✓	-
IDL	✓	✓	✓	✓	-	-	✓	✓
LDL	✓	✓	✓	✓	-	-	✓	✓
LDL-1 to -6	✓	✓	✓	✓	-	-	✓	✓
HDL	✓	✓	✓	✓	✓	✓	-	-
HDL-1 to -4	✓	✓	✓	✓	✓	✓	-	-

Lipoprotein turnover

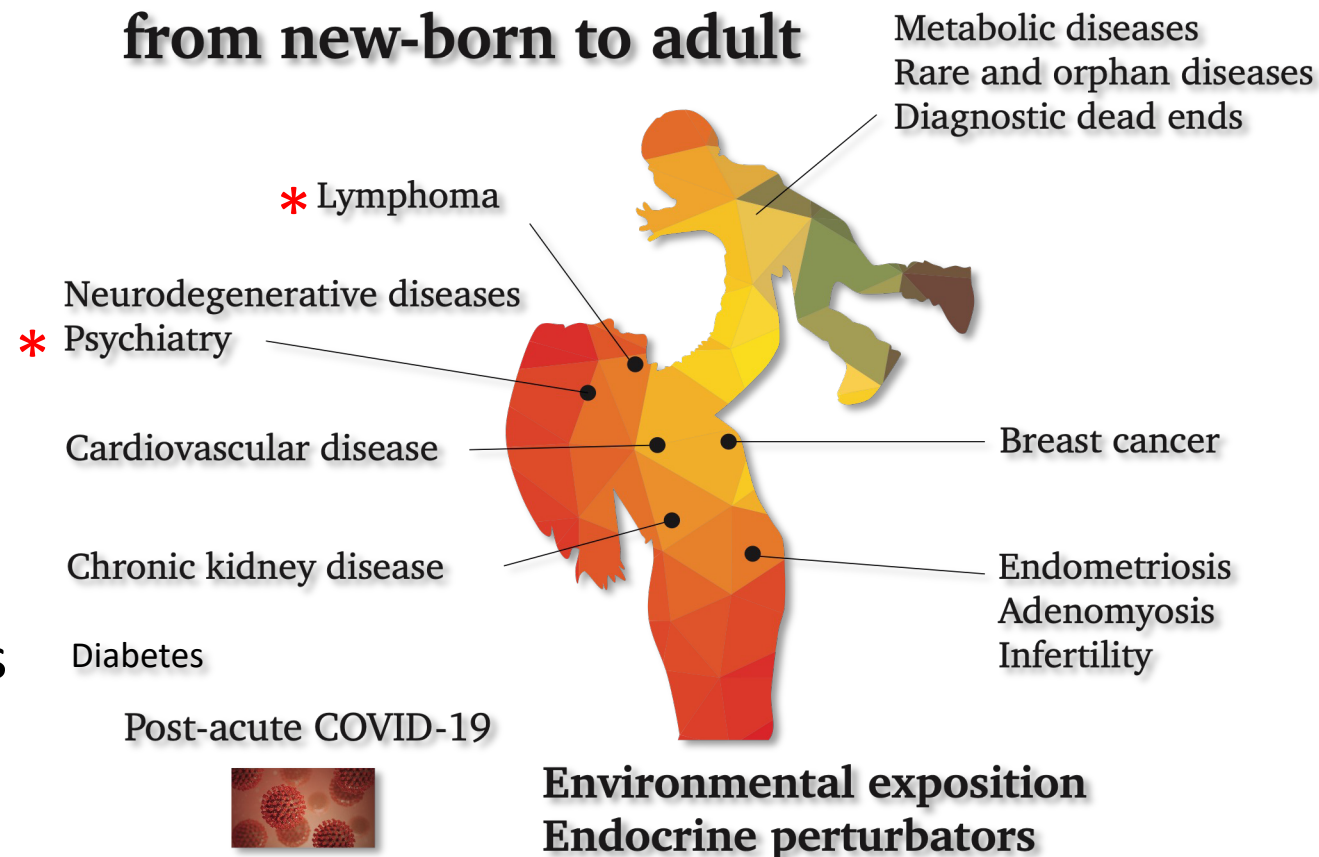


HL : hepatic lipase
LCAT :
lecithine:cholesterol
acyltransferase
RLP-C : remnant
lipoprotein
cholesterol

HUMAN HEALTH PROJECTS @ Univ Paris Cité

- Longitudinal follow-up
- Cohort study
- Patient stratification
- Diagnostic / Prognosis
- Monitoring the efficiency of treatments

from new-born to adult



Mental Health



MS & NMR

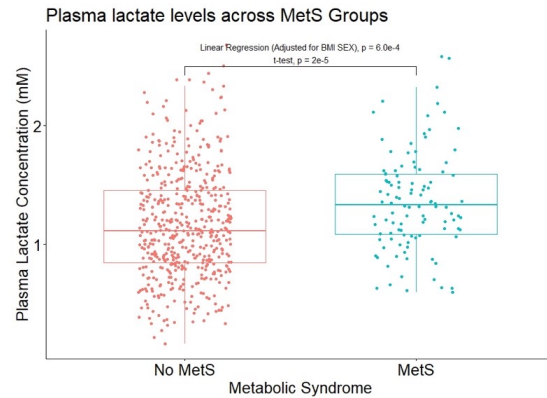
MS-based Metabolomics

Blood biomarkers

➤ Mitochondrial biomarkers and metabolic syndrome in bipolar disorder

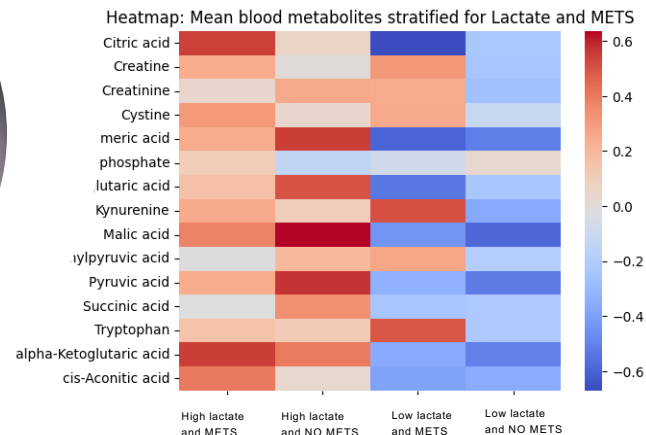
• Biological mechanisms involved in Metabolic syndrome

Ophélie Godin



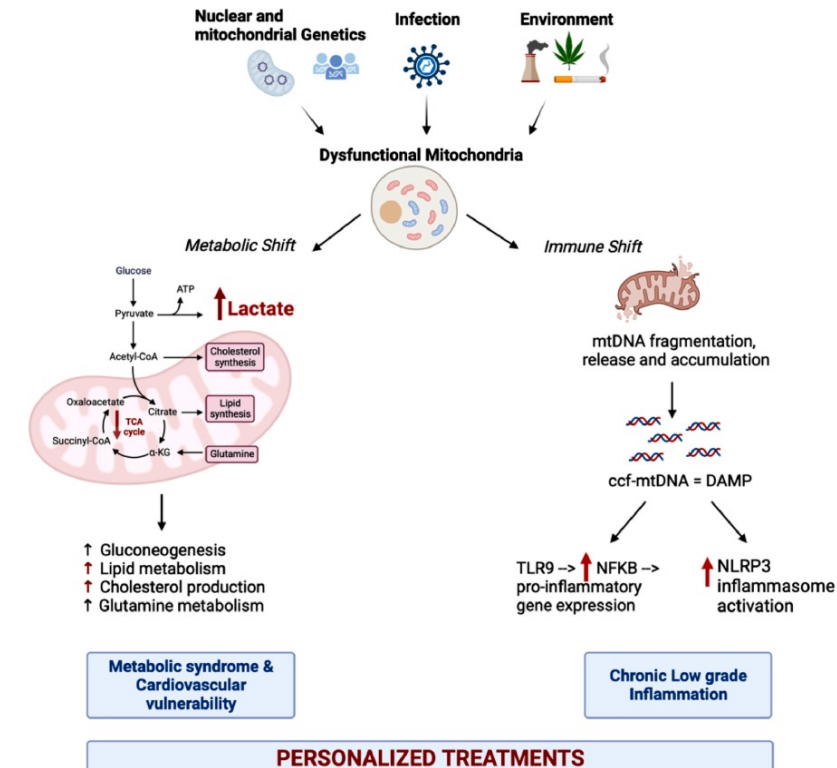
➔ Lactate is associated with MetS

Marion Leboyer



➔ Using mitochondrial targeted MS-metabolomics analysis
Different metabolite profile identified

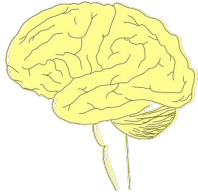
- MetS + Lactate + : citric acid & α KG acid
- MetS + Lactate - : kynurenine & tryptophan



FACE-BD cohort

Zachos, Choi, Godin, et al. *Psychiatry Res.* 2024;339:116063

NMR-based Metabolomics



- ❑ Brain & lipids : richest organ in terms of lipid & cholesterol content
- ❑ Lipoproteins : vectors of lipids in the body

- ❖ Metabolites
- ❖ Inflammatory biomarkers
- ❖ Lipoproteins

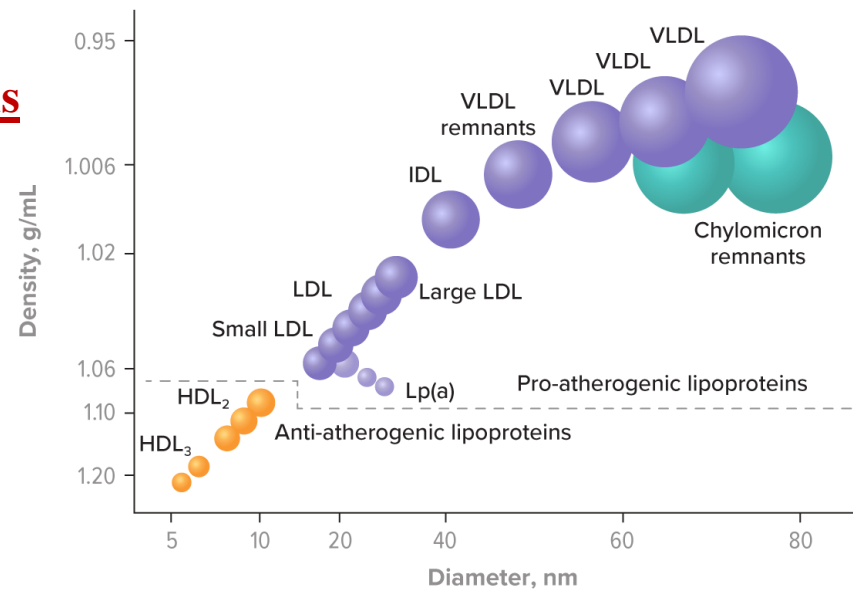
➤ Only 4 clinical lipid parameters
(namely HDL-Chol, LDL-Chol, Total Chol, Triglycerides)

➤ Deciphering lipoproteome at a level never seen before : **108 additional parameters !**

Lipoprotein subclass analysis



IVDr NMR technology



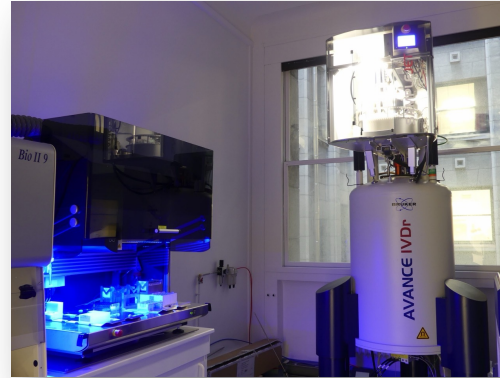
VLDL 1 to 6
LDL 1 to 6
HLD 1 to 4
Apo A1, A2, B100
...

TPTG	HDA2	L5CH	H3PL
TPCH	VLAB	L6CH	H4PL
LDCH	IDAB	L1FC	H1A1
HDCH	LDAB	L2FC	H2A1
TPA1	V1TG	L3FC	H3A1
TPA2	V2TG	L4FC	H4A1
TPAB	V3TG	L5FC	H1A2
TPTG	V4TG	L6FC	H2A2
ABA1	V5TG	L1PL	H3A2
TBPN	V1CH	L2PL	H4A2
VLPN	V2CH	L3PL	
IDPN	V3CH	L4PL	
LDPN	V4CH	L5PL	
L1PN	V5CH	L6PL	
L2PN	V1FC	L1AB	
L3PN	V2FC	L2AB	
L4PN	V3FC	L3AB	
L5PN	V4FC	L4AB	
L6PN	V5FC	L5AB	
VLTG	V1PL	L6AB	
IDTG	V2PL	H1TG	
LDTG	V3PL	H2TG	
HDTG	V4PL	H3TG	
VLCH	V5PL	H4TG	
IDCH	L1TG	H1CH	
VLFC	L2TG	H2CH	
IDFC	L3TG	H3CH	
LDFC	L4TG	H4CH	
HDFC	L5TG	H1FC	
VLPL	L6TG	H2FC	
IDPL	L1CH	H3FC	
LDPL	L2CH	H4FC	
HDPL	L3CH	H1PL	
HDA1	L4CH	H2PL	

NMR-based Metabolomics

IVDr NMR : In Vitro Diagnosis for research

- ✓ High reproducibility
- ✓ Simple sample preparation
- ✓ High-throughput analysis
- ✓ Absolute quantification



Bruker Corporation (NASDAQ: BRKR) more than 11,000 employees worldwide

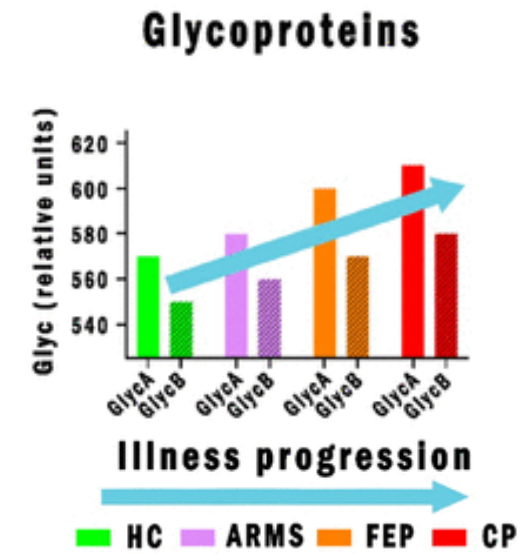
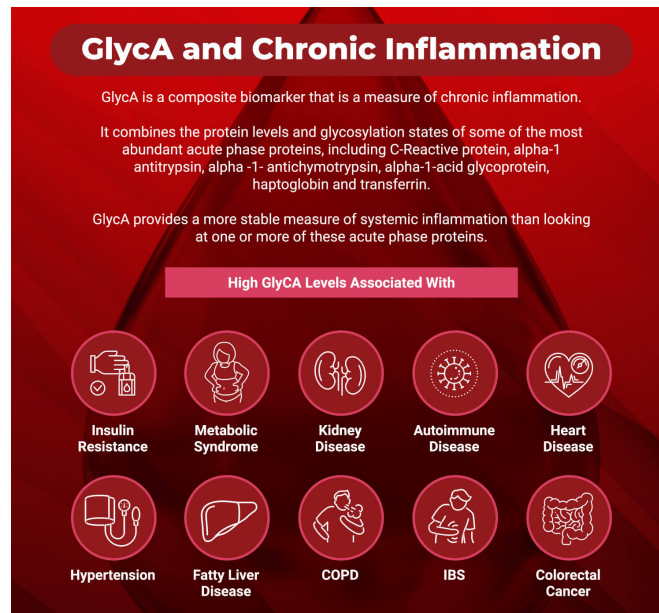
Blood biomarkers

➤ 323 metabolic measures (158 pure biomarkers)

- ✓ Metabolites
- ✓ Inflammatory biomarkers
- ✓ Lipoproteins

➤ Psychosis

GlycA : a novel biomarker of Low-grade chronic Inflammation



Healthy control

Critical Period (>2 episodes)

First Episode of Psychosis

At-Risk Mental State

Gradual Increase in Inflammation-Linked Glycoproteins and a Proatherogenic Lipoprotein Profile in the Early Stages of Psychosis as Characterized by ^1H NMR Blood Analysis. J Proteome Res. 2023

* ➤ Psychosis : NMR-based Metabolomics

Ariel Frajerman

Boris Chaumette

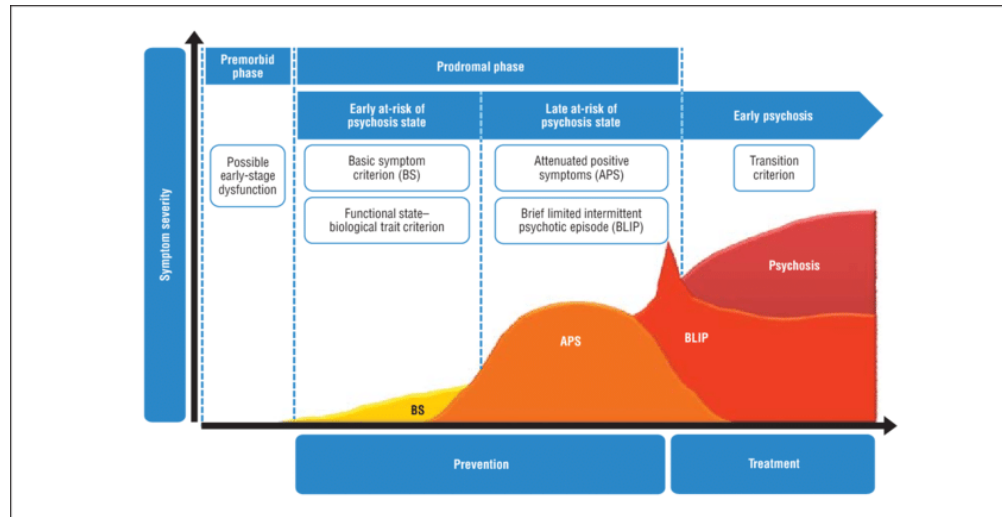


UHR patients : ultra high risk of transition to psychosis
15% within the first year from the diagnosis; 25% after three years

➤ Objectives : Identify blood metabolomic biomarkers for psychotic conversion

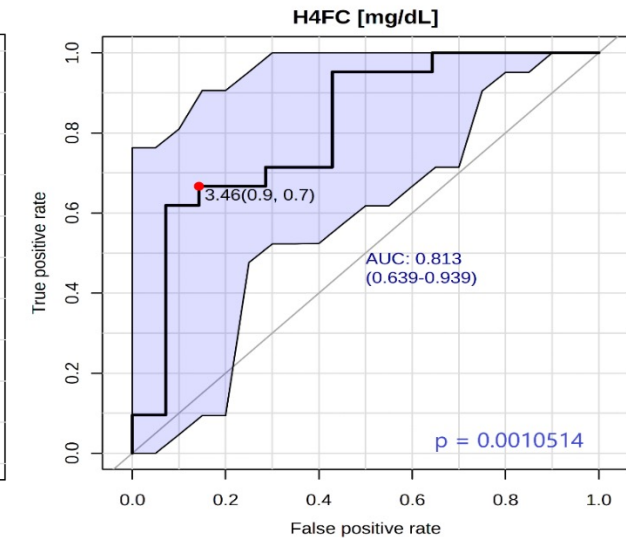
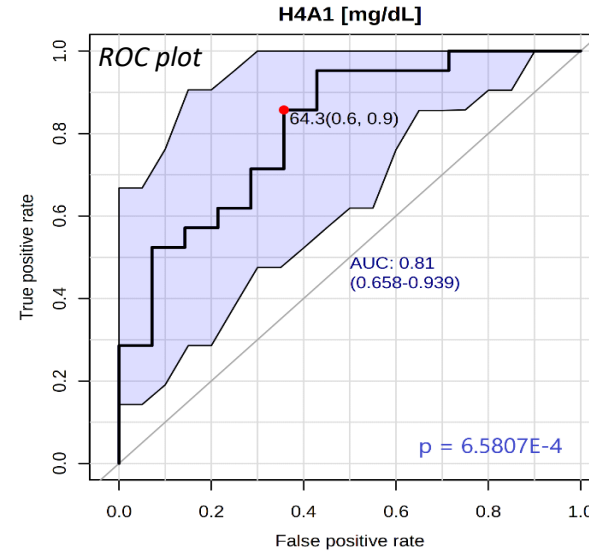
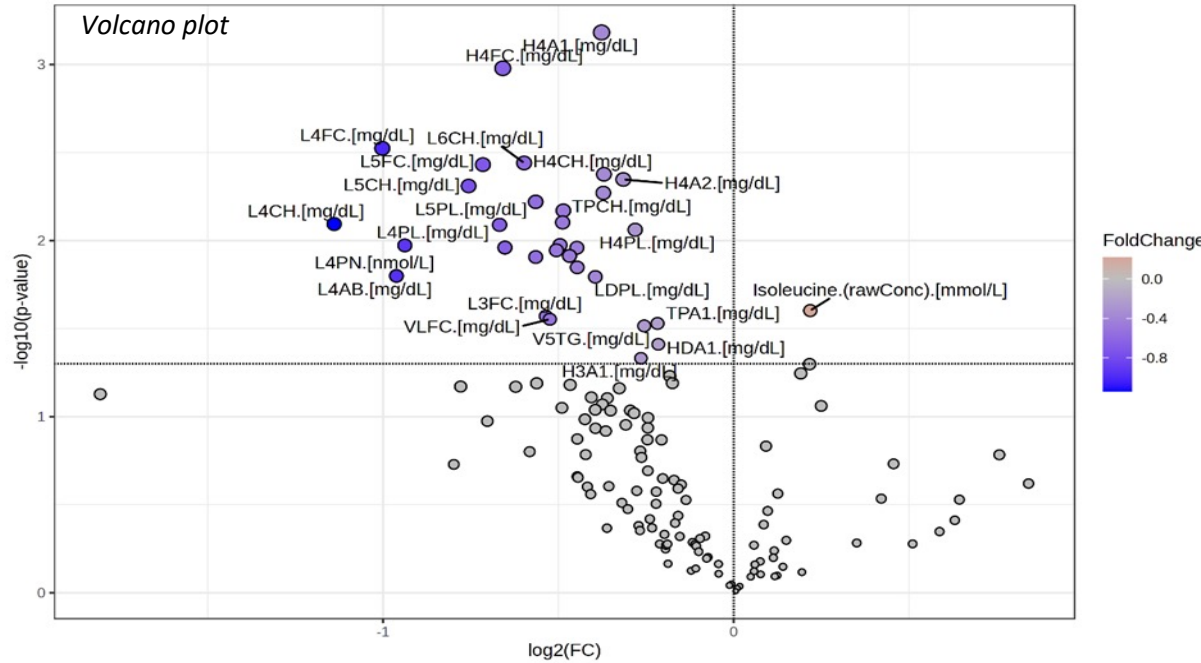
In First Episode of Psychosis (FEP), abnormalities have been identified previously in cholesterol lipoproteins (HDL-Chol, LDL-Chol) & triglycerides

➤ **A pilot study** on ICAAR cohort => plasmas of 35 UHR patients
14 converters (UHR-C), 21 non-converters (UHR-NC) at inclusion



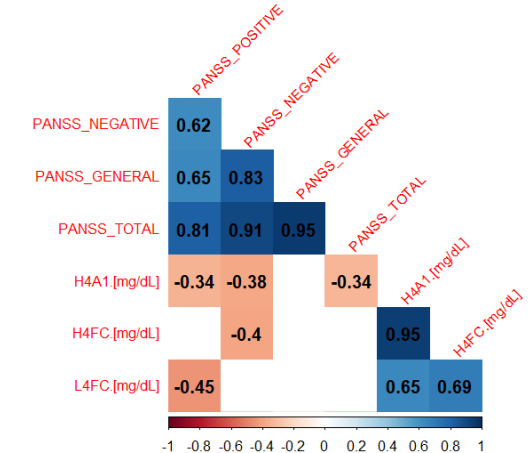
Characteristic	UHR-C, N = 14	UHR-NC, N = 21	p-value
Sex			>0.9
Women	6 (43%)	9 (43%)	
Men	8 (57%)	12 (57%)	
Age	20.71 ± 2.95	21.33 ± 4.08	0.6
BMI	22.31 ± 4.17	22.99 ± 4.14	0.6
Antipsychotic	0 (0%)	1 (5%)	>0.9
PANSS Total	80.00 ± 18.88	57.10 ± 12.71	<0.001
PANSS General	44.07 ± 10.34	34.95 ± 8.02	0.010
PANSS Positive	18.79 ± 5.12	9.86 ± 3.24	<0.001
PANSS Negative	17.14 ± 6.83	12.29 ± 4.82	0.031

Results



- Global HDL-Chol is not relevant
- Low-level of HDL-4 predicted psychotic conversion

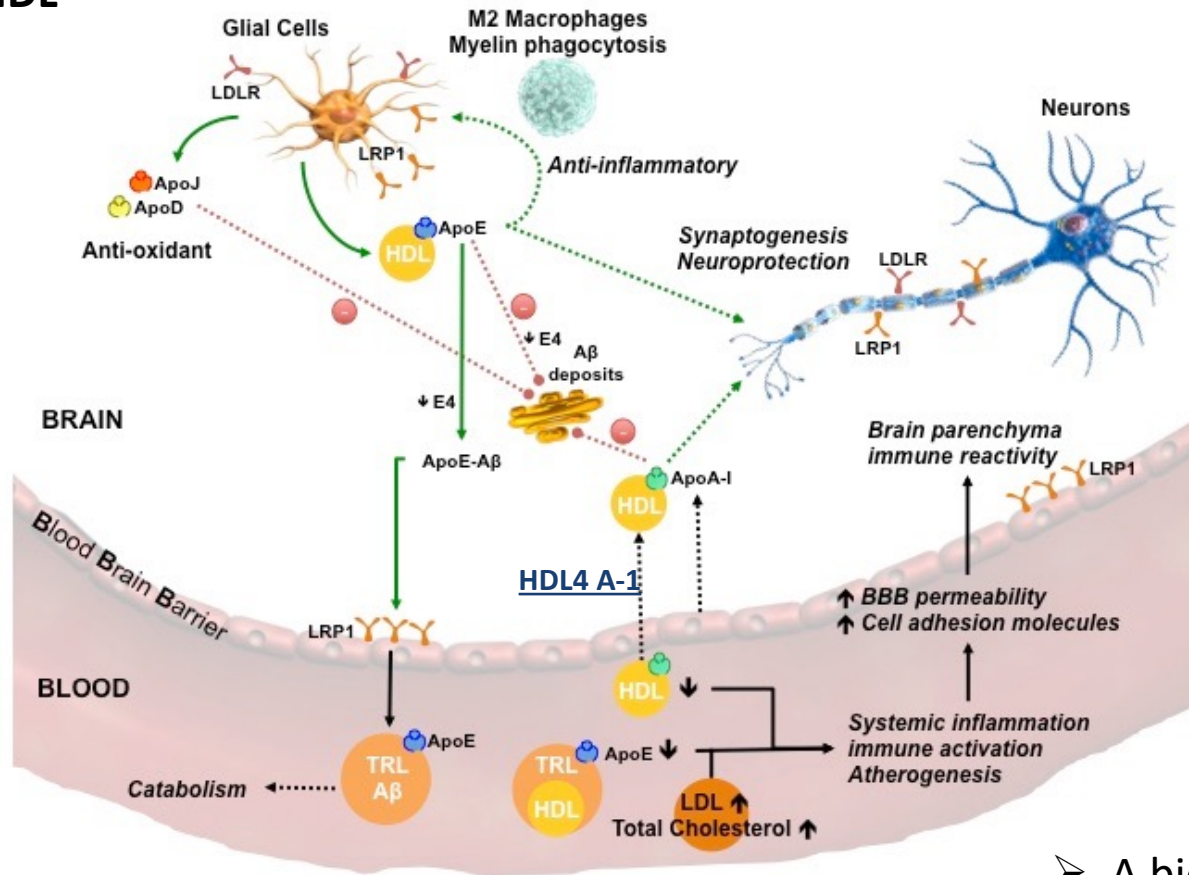
➤ Correlated to PANSS



➤ Interpretation

Why we found HDL-4 ?

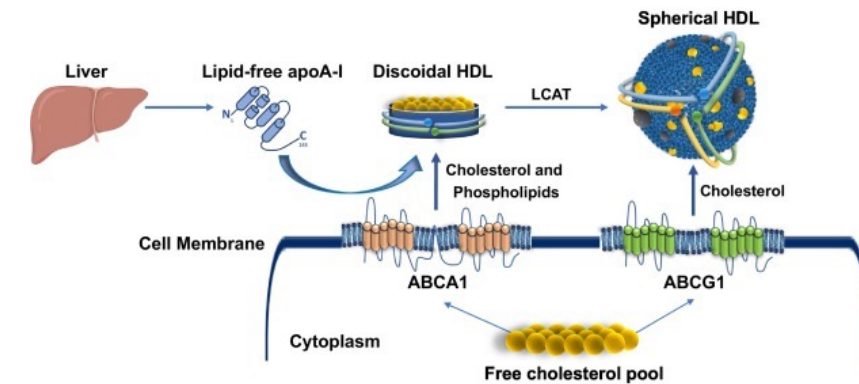
- 'smallest' HDL
- ApoA-1



Sena, et al. Plasma Lipoproteins in Brain Inflammatory and Neurodegenerative Diseases'. Lipoproteins - Role in Health and Diseases, InTech, 2012

➤ Discoidal HDL

HDL-4 apoA1 with Free Cholesterol



Ong KL, et al. HDL maturation and remodelling. Biochim Biophys Acta Mol Cell Biol Lipids. 2022

- A biomarker found recently in serum of AD patients

Berezhnoy G, Laske C, Trautwein C. Quantitative NMR-Based Lipoprotein Analysis Identifies Elevated HDL-4 and Triglycerides in the Serum of Alzheimer's Disease Patients. Int J Mol Sci. 2022

* ➤ Diffuse Large B-Cell Lymphoma : A common aggressive lymphoma



BLOOD

➤ Although its treatment outcome has improved, 40% of patients will be refractory or relapse, and die upon 1st-Line therapy

➤ Objectives :

Early detection of patients at high-risk of death and/or relapse



Collaboration:

Dr V. Baud

Faculté de Pharmacie, Univ. Paris Cité



Pr C. Thieblemont

Hôpital Saint-Louis, Paris

Cohort REMARC



Trial study started including patients in 2010 and was a double-blind, international, multicenter, randomized phase II study

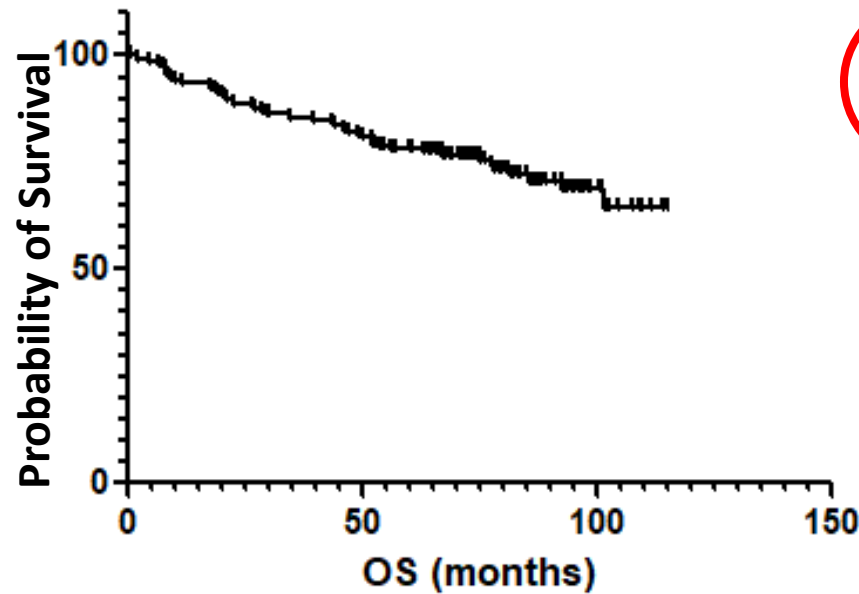
- 154 patients
- Blood sample collected at diagnosis of lymphoma
- Before 1st-line therapy

DLBCL

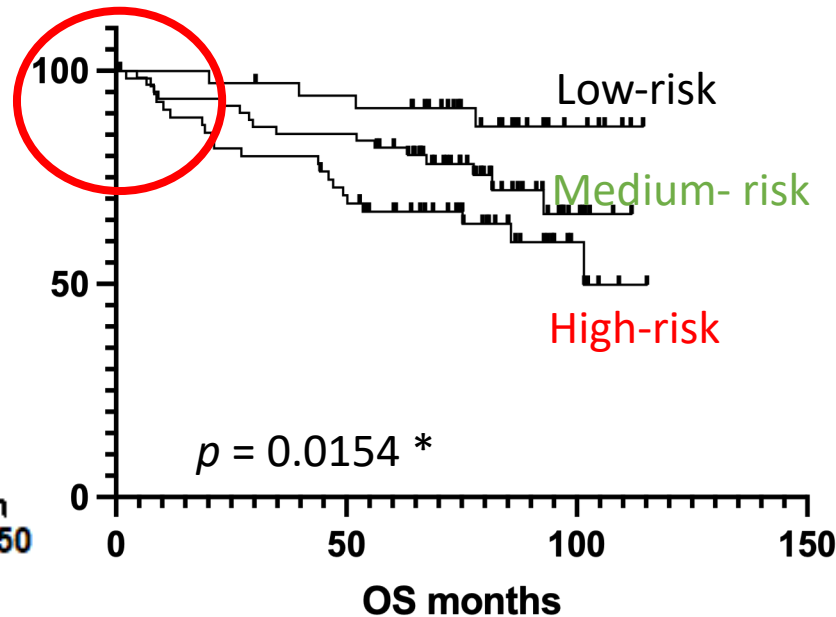
➤ Overall Survival (OS)

Length of time from the date of diagnosis that patients diagnosed with the disease are still alive.

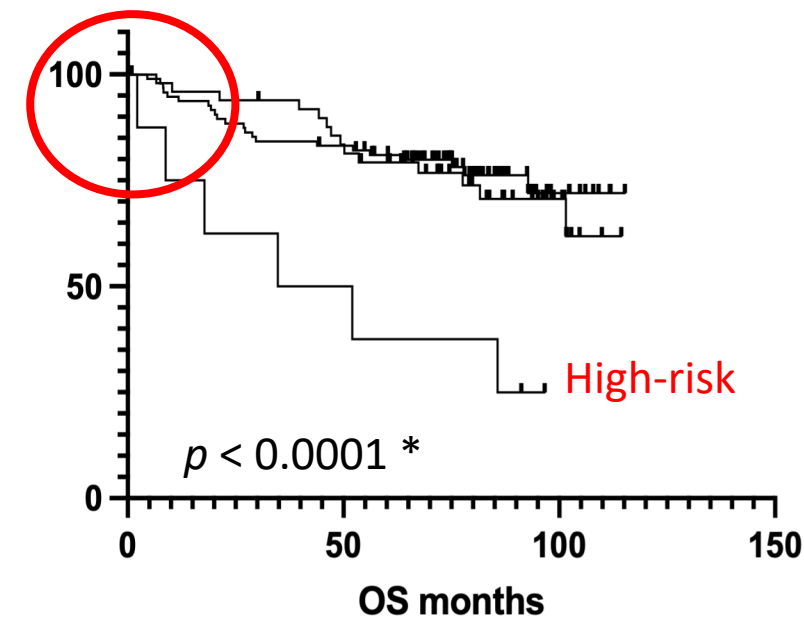
Cohort evolution



IPI : international scoring index



NMR-based scoring



Confidential biomarker

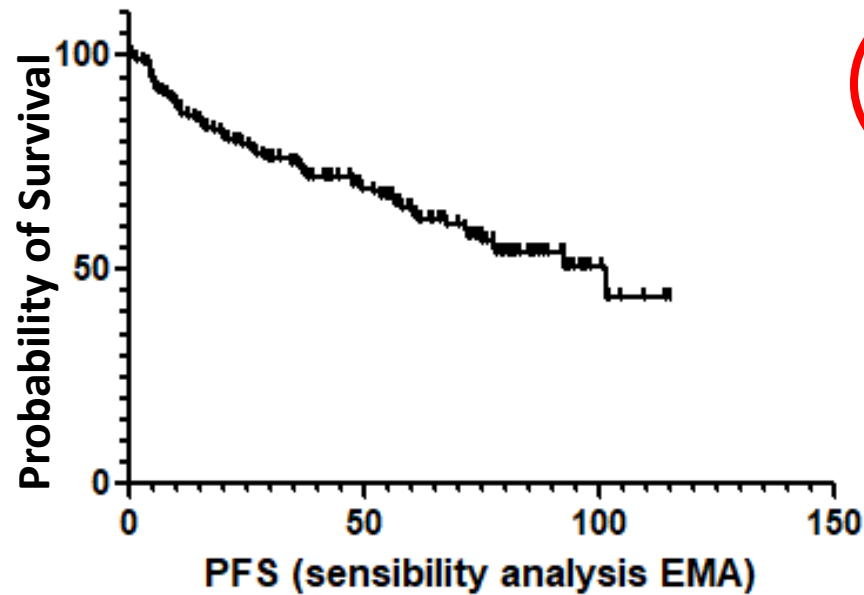
* Log-rank (Mantel-Cox) Test

DLBCL

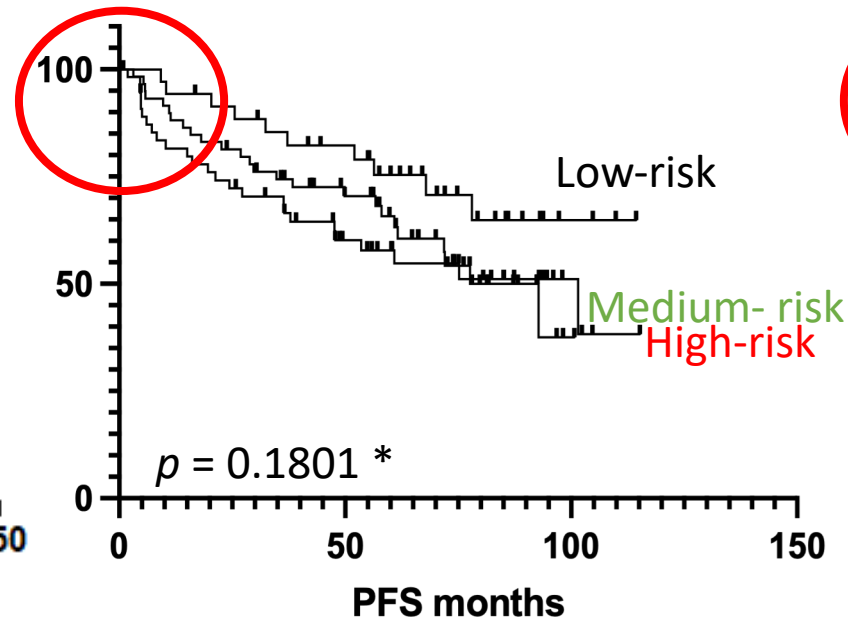
➤ Profile free-survival (PFS) curves

Length of time during and after the treatment of a disease that a patient lives with the disease but it does not get worse.

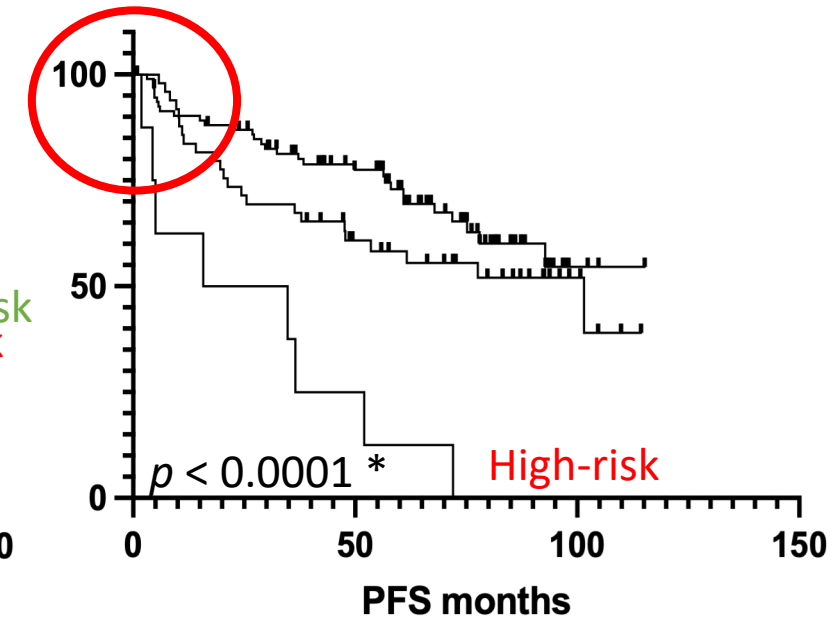
Cohort evolution



IPI : international scoring index



NMR-based scoring



Confidential biomarker

* Log-rank (Mantel-Cox) Test

DISCUSSION OPPORTUNITIES



Key points to make good metabolomics ?

- IMPORTANCE OF **LARGE-SCALE STUDY**
- ESTABLISH **OMICS DATABASES**
- AGREED UPON **STANDARDS**
- HOLISTIC APPROACH (**NON-TARGETED** at first intention)

